

AWS Website Deployment on Amazon EC2

Deploying a Secure Static Website Using Ubuntu Linux, Nginx & Route 53

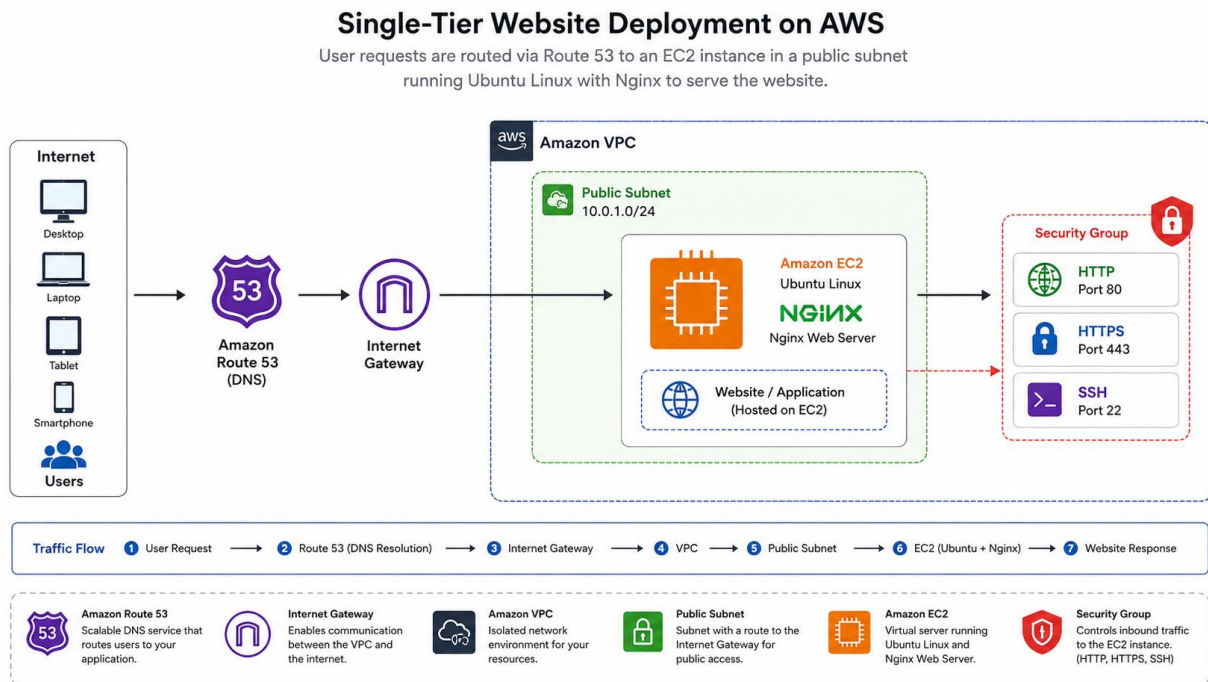
Prepared by:

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Project Overview

This project demonstrates the successful deployment of a secure static website on Amazon Web Services (AWS). The website was hosted on an Amazon EC2 instance running Ubuntu Linux with Nginx configured as the web server. Amazon Route 53 was used for DNS configuration, while AWS Security Groups were implemented to secure network access.

The project showcases practical experience in cloud infrastructure deployment, Linux server administration, web server configuration, DNS management, and AWS networking.



Technologies Used

Cloud Platform

- Amazon Web Services (AWS)

AWS Services

- Amazon EC2
- Amazon VPC
- Amazon Route 53
- AWS Security Groups

Operating System

- Ubuntu Linux

Web Server

- Nginx

Tools

- SSH
- Linux Terminal

Project Objectives

- Deploy a static website on Amazon EC2.
- Configure Ubuntu Linux for web hosting.
- Install and configure Nginx as the web server.
- Configure DNS using Amazon Route 53.
- Secure the server using AWS Security Groups.
- Verify that the website is publicly accessible.

Implementation Process

Step 1 – Launch an Amazon EC2 Instance

Created and configured an Ubuntu EC2 instance to host the website.

EC2 > Instances

Instances (1/1) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set [Find Instance by attribute or tag \(case-sensitive\)](#)

☒	Name 🔗	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public
☒	space-rise-we...	i-07ec0bf3d4646784f	Running 🔍	t2.medium	Initializing 🕒	us-east-1a	-	-

i-07ec0bf3d4646784f (space-rise-webserver) [🔧](#) [▼](#)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ **Instance summary** [Info](#)

Instance ID 🔗 i-07ec0bf3d4646784f	Public IPv4 address -	Private IPv4 addresses 🔗 192.168.1.10
IPv6 address	Instance state 🔍 Running	Public DNS

Step 2 – Configure Security Groups

Configured inbound rules to allow:

- SSH (Port 22)
- HTTP (Port 80)
- HTTPS (Port 443)

EC2 > Security Groups > sg-08533057c4f39dd3e > Edit inbound rules

Edit inbound rules [Info](#)
Inbound rules control the incoming traffic that's allowed to reach the instance.

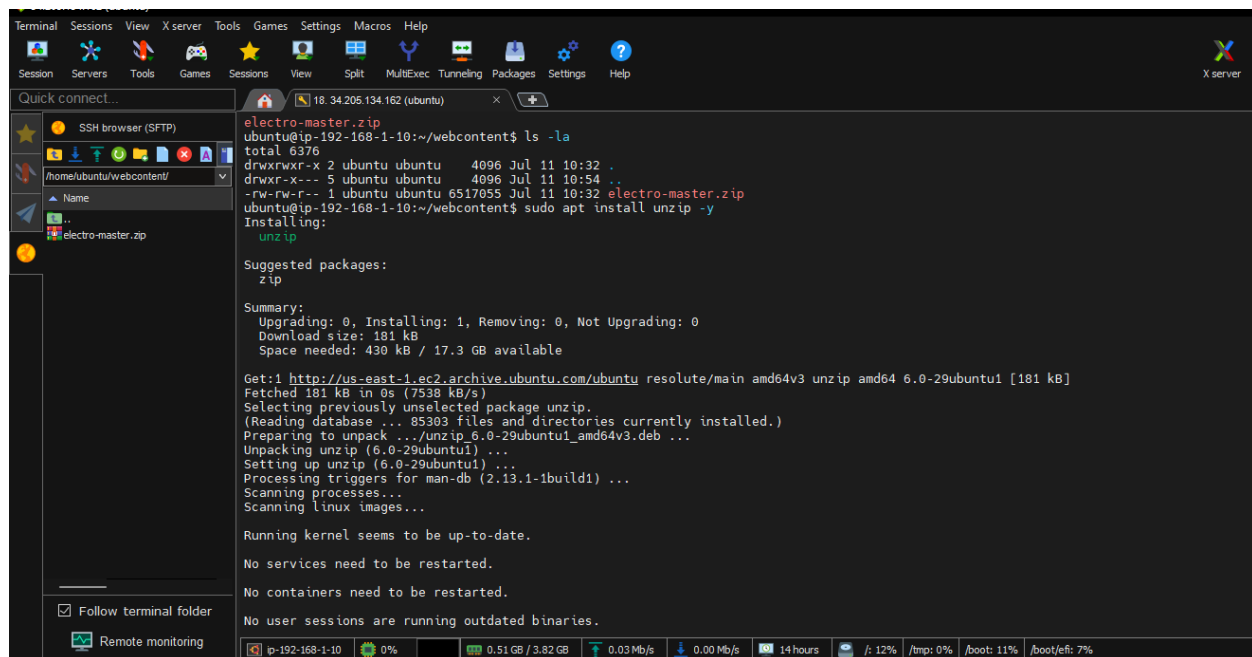
Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0df510b60cd3b2107	SSH	TCP	22	Cus... 🔍	102.90.124.190/32 ✕	Delete
sgr-049df70f40b63b76c	HTTP	TCP	80	Cus... 🔍	102.90.124.190/32 ✕ this rule is for only me	Delete
sgr-04b2ce83b0ee9d119	HTTPS	TCP	443	Cus... 🔍	102.90.124.190/32 ✕ 0.0.0.0/0 ✕	Delete

[Add rule](#)

⚠️ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. [✕](#)

Step 3 – Connect to the Server

Connected securely to the EC2 instance using SSH for remote administration.



Step 4 – Configure Ubuntu Linux

Updated the server packages and prepared the Linux environment for deployment.

Commands used:

```
sudo apt update
```

```
sudo apt upgrade
```

Step 5 – Install Nginx

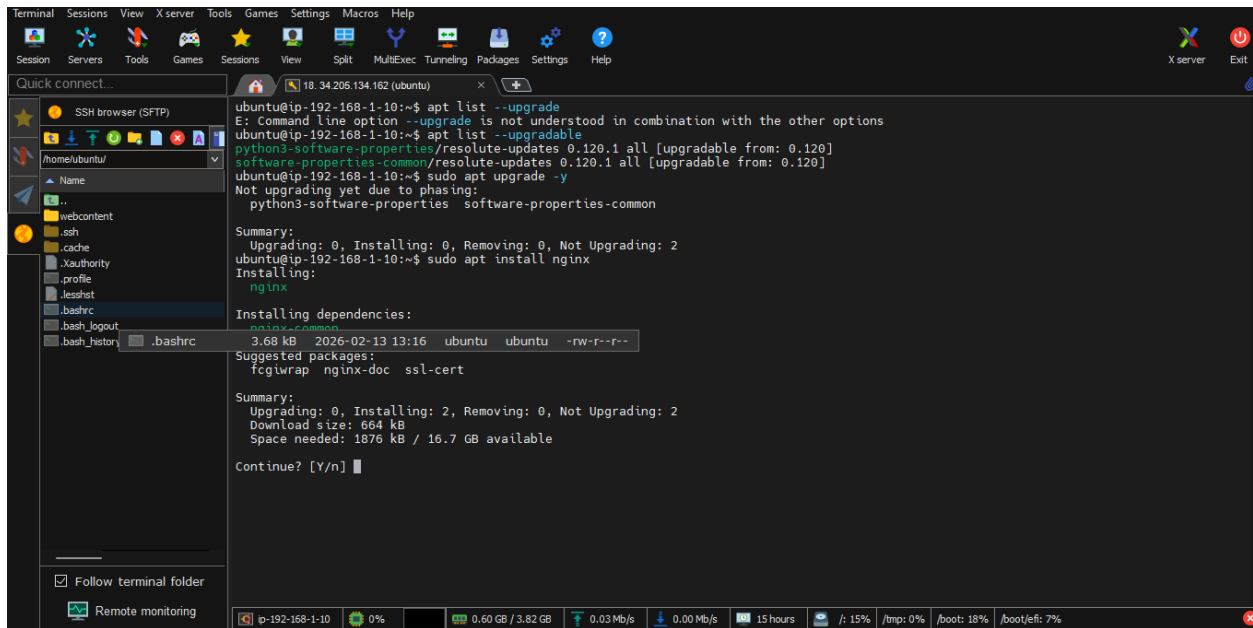
Installed and configured Nginx to serve the website.

Commands used:

```
sudo apt install nginx
```

```
sudo systemctl start nginx
```

```
sudo systemctl enable nginx
```



```
ubuntu@ip-192-168-1-10:~$ apt list --upgrade
E: Command line option --upgrade is not understood in combination with the other options
ubuntu@ip-192-168-1-10:~$ apt list --upgradable
python3-software-properties/resolute-updates 0.120.1 all [upgradable from: 0.120]
software-properties-common/resolute-updates 0.120.1 all [upgradable from: 0.120]
ubuntu@ip-192-168-1-10:~$ sudo apt upgrade -y
Not upgrading yet due to phasing:
python3-software-properties software-properties-common

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
ubuntu@ip-192-168-1-10:~$ sudo apt install nginx
Installing:
nginx

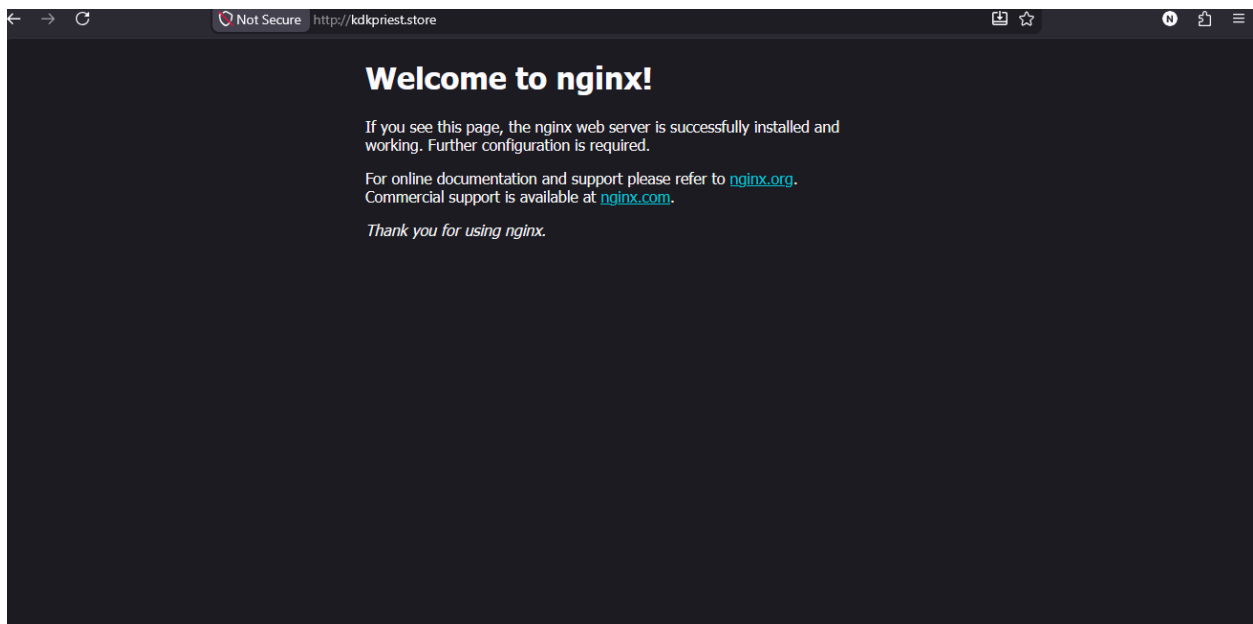
Installing dependencies:
3.68 kB 2026-02-13 13:16 ubuntu ubuntu -rw-r--r--
Suggested packages:
fcgiwrap nginx-doc ssl-cert

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2
Download size: 664 kB
Space needed: 1876 kB / 16.7 GB available

Continue? [Y/n]
```

Step 6 – Deploy the Website

Uploaded the website files to the Nginx web root directory and verified successful deployment.



Step 7 – Configure Route 53

Configured DNS records to connect the domain name to the EC2 instance.

VPC > Route tables > rtb-063f3e87e0f37f3d8 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.1.0/28	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute

Add route

Cancel Preview Save changes

Project Outcome

The static website was successfully deployed on Amazon EC2 using Ubuntu Linux and Nginx. DNS resolution was configured using Amazon Route 53, and the infrastructure was secured using AWS Security Groups. The deployment demonstrates foundational cloud engineering skills and best practices for hosting static websites on AWS.

Skills Demonstrated

- AWS Cloud Computing
- Amazon EC2
- Amazon VPC
- Amazon Route 53
- Ubuntu Linux
- Linux Server Administration
- Nginx
- SSH
- DNS Configuration
- Website Deployment
- Cloud Networking
- Cloud Security

Deliverables

- ✓ Amazon EC2 Instance Deployment
- ✓ Ubuntu Linux Server Configuration
- ✓ Nginx Installation and Configuration
- ✓ Static Website Deployment
- ✓ Route 53 DNS Configuration
- ✓ AWS Security Group Configuration
- ✓ SSH Remote Administration
- ✓ Functional Website Verification

Why Choose My Service?

I deploy secure, reliable, and well configured static websites on AWS using industry best practices. Every deployment is carefully configured with security, performance, and scalability in mind. Whether you need a personal portfolio, business website, or landing page, I can help you host it efficiently on Amazon Web Services.

Contact

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